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B.Tech. Degree VI Semester Special Supplementary Examination November 2022

ME 19-205-0602 MACHINE DESIGN I
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

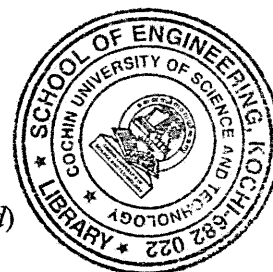
Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the customers' need, formulate the problem and draw the design specifications.
 CO2: Understand component behaviour subjected to loads and identify the failure criteria.
 CO3: Apply the concepts of stress analysis, theories of failure and material science to analyse, design and/or select commonly used machine components.
 CO4: Analyse the stresses and strains induced in a machine element.
 CO5: Design machine components using theories of failure.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

(Use of Design Data Handbook is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

**PART A**

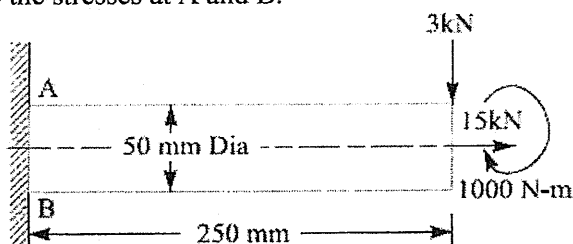
(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What is meant by stress concentration?	3	3	L1	2	1
(b) State and explain the term 'Resilience'.	3	3	L2	1	1
(c) Give at least three practical applications of couplings.	3	3	L2	1	1,2
(d) What is meant by preloading of bolts?	3	3	L1	2	1
(e) What are various modes in which a riveted joint may fail?	3	3	L1	2	1,2
(f) Explain A.M. Wahl's factor and state its importance in designing helical springs.	3	3	L2	1	1,2
(g) Discuss the advantages of welded joints in comparison to riveted joints.	3	3	L2	1	1
(h) Explain the term: 'critical speed of shaft'.	3	3	L2	2	1

PART B

(4 × 12 = 48)

- II. A shaft, as shown in figure, is subjected to a bending load of 3 kN, pure torque of 1000 N-m and an axial pulling force of 15 kN. Calculate the stresses at A and B.



OR

(P.T.O.)

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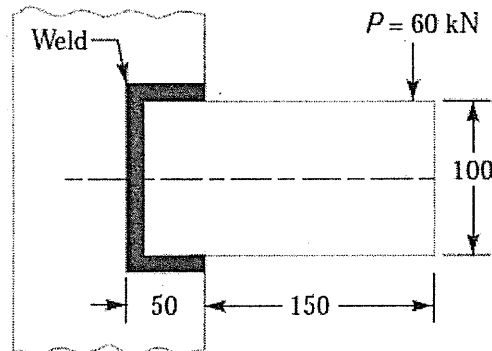
		Marks	BL	CO	PO
III.	A mild steel shaft of 50 mm diameter is subjected to a bending moment of 2000 N-m and a torque T. If the yield point of the steel in tension is 200 MPa, find the maximum value of this torque without causing yielding of the shaft according to: (i) The maximum principal stress theory (ii) The maximum shear stress theory (iii) The maximum distortion strain energy theory of yielding.	12	L3	5	1,2
IV.	Design a gib and cotter joint to carry a maximum load of 35 kN. Assuming that the gib, cotter and rod are of same material and have the following allowable stresses : $\sigma_t = 20 \text{ MPa}$; $\tau = 15 \text{ MPa}$; and $\sigma_c = 50 \text{ MPa}$ OR	12	L4	3	1,2,3
V.	Design a cast iron protective type flange coupling to transmit 15 kW at 900 rpm from an electric motor to a compressor. The service factor may be assumed as 1.35. The following permissible stresses may be used : Shear stress for shaft, bolt and key material = 40 MPa Crushing stress for bolt and key = 80 MPa Shear stress for cast iron = 8 MPa Draw a neat sketch of the coupling.	12	L4	3	1,2
VI.	Design the longitudinal joint for a 1.25 m diameter steam boiler to carry a steam pressure of 2.5 MPa. The ultimate strength of the boiler plate may be assumed as 420 MPa, crushing strength as 650 MPa and shear strength as 300 MPa. Take the joint efficiency as 80%. Sketch the joint with all the dimensions. Adopt the suitable factor of safety. OR	12	L3	3	1,2,3
VII.	Design a helical spring for a spring loaded safety valve (Rams bottom safety valve) for the following conditions : Diameter of valve seat = 65 mm Operating pressure = 0.7 MPa Maximum pressure when the valve blows off freely = 0.75 MPa Maximum lift of the valve when the pressure rises from 0.7 to 0.75 MPa = 3.5 mm Maximum allowable stress = 550 MPa Modulus of rigidity = 84 GPa Spring index = 6	12	L4	5	1,2,3

(Continued)

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VIII.

A rectangular steel plate is welded as a cantilever to a vertical column and supports a single concentrated load P , as shown in figure. Determine the weld size if shear stress in the same is not to exceed 140 MPa.



OR

IX.

A shaft is supported by two bearings placed 1 m apart. A 600 mm diameter pulley is mounted at a distance of 300 mm to the right of left hand bearing and this drives a pulley directly below it with the help of belt having maximum tension of 2.25 kN. Another pulley 400 mm diameter is placed 200 mm to the left of right hand bearing and is driven with the help of electric motor and belt, which is placed horizontally to the right. The angle of contact for both the pulleys is 180° and $\mu = 0.24$. Determine the suitable diameter for a solid shaft, allowing working stress of 63 MPa in tension and 42 MPa in shear for the material of shaft. Assume that the torque on one pulley is equal to that on the other pulley.

Bloom's Taxonomy Levels

L1 = 7.5%, L2 = 12.5%, L3 = 20.0%, L4 = 50.0%, L5 = 10.0%.

Marks	BL	CO	PO
12	L4	3	1,2,3

12	L5	4	1,2,3
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